

Teaching Tech Together

Cognitive Load



Greg Wilson
<https://third-bit.com>

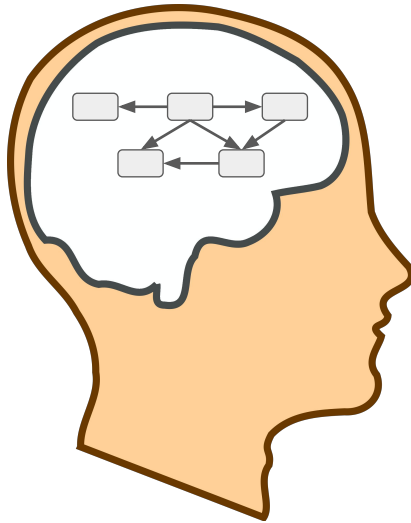


All art from openclipart.org (open license)

Learning Objectives

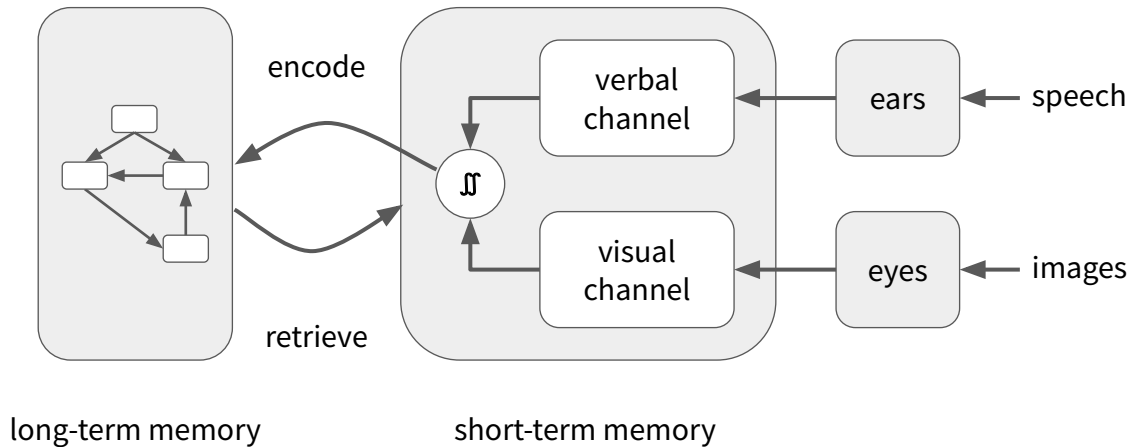
1. Explain the difference between long-term and short-term memory and describe the retrieval/encoding cycle.
2. Describe how visual and auditory information are processed, integrated, and stored.
3. Describe the cognitive processing involved in reading and summarize Benner et al's findings about learning modes (visual + speech, visual + text, visual + speech + text).
4. Define the three types of cognitive load and correctly identify examples of each.
5. Explain two implications of cognitive load theory for lesson design and delivery.
6. Explain what a Parsons Problem is and how its effectiveness can be grounded in cognitive load theory.
7. Create Parsons Problems and explain when and how they should or shouldn't be used.

What's Going On In There?



We've been talking about mental models as if they were real things, but what actually goes on in a learner's brain when they're learning? The short answer is that we don't know; the longer answer is that we do know a lot more than we used to, and have some pretty good models that can help us design and deliver lessons more effectively.

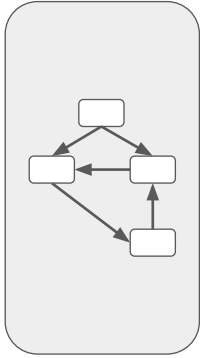
Cognitive Architecture (Simplified)



This is a simplified model of human cognitive architecture. There's a lot going on here, so we're going to take it in pieces.

Description: a diagram showing long-term memory, short-term memory, and speech and images coming into the ears and eyes respectively. The next few slides describe this diagram in more detail.

Two Kinds of Memory



long-term memory

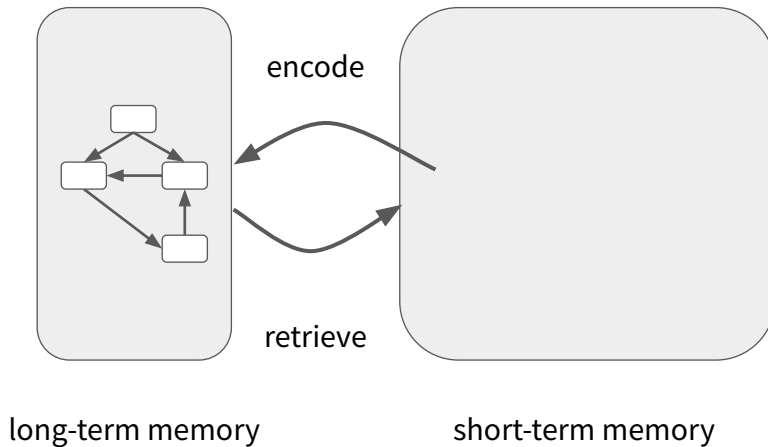


short-term memory

The core of this model is the fact that you have two kinds of memory. *Long-term memory* is your basement: it stores things more or less permanently, but your consciousness can't access it directly. Instead, you rely on your *short-term memory*, which is your desktop.

Description: a box on the left labeled "long-term memory" that contains a graph meant to suggest a mental model. An empty box in the middle is labeled "short-term memory".

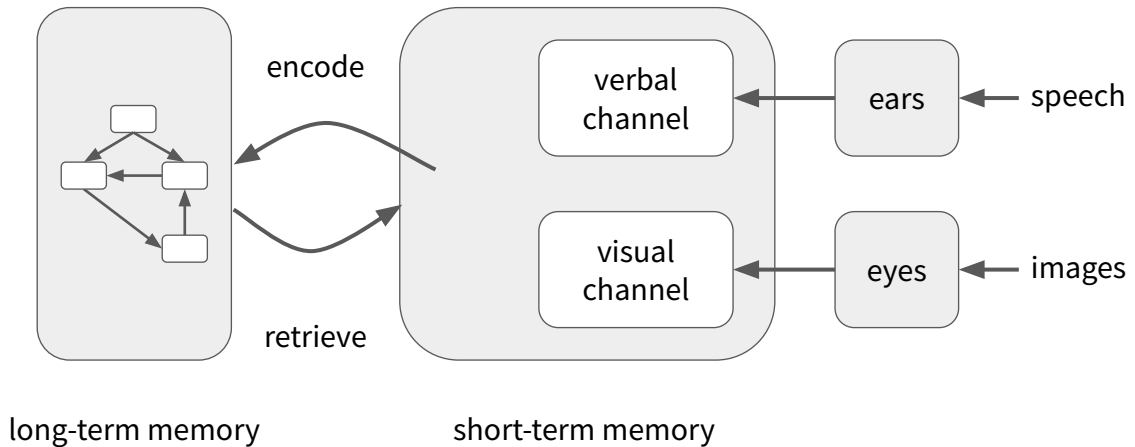
Encoding and Retrieving



When you need something, your brain *retrieves* it from long-term memory and puts it in short-term memory. New information that arrives in short-term memory has to be *encoded* to be stored in long-term memory. If that information isn't encoded and stored, it's not remembered and learning hasn't taken place.

Description: the same boxes labeled "short-term memory" and "long-term memory", now joined by a pair of arrows. The arrow from long-term memory to short-term memory is labeled "retrieve", and the arrow from short-term memory to long-term memory is labeled "encode".

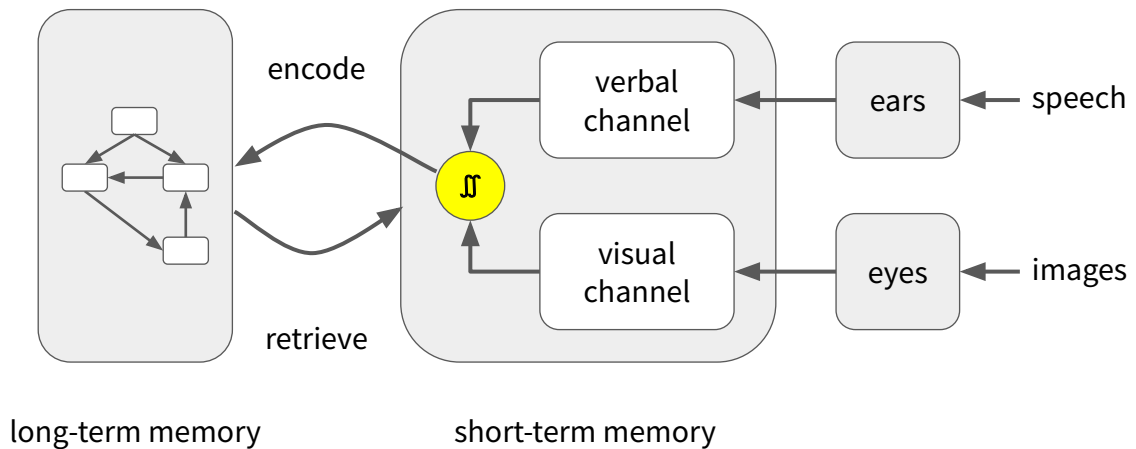
Verbal and Visual Channels



Information gets into short-term memory primarily through your *verbal channel* (for speech) and *visual channel* (for images). A more complete model would also include your sense of touch, and your ability to smell and taste things, but we'll ignore those for now.

Description: the same diagram of long-term and short-term memory as before. On the right, there are now two small boxes labeled "ears" and "eyes". An arrow labeled "speech" goes into the "ears" box, and another arrow goes from the "ears" box to a small box labeled "verbal channel" inside the box labeled "short-term memory". Similarly, an arrow labeled "images" goes into the "eyes" box, and another arrow goes from "eyes" to a small box labeled "visual channel" inside the "short-term memory" box.

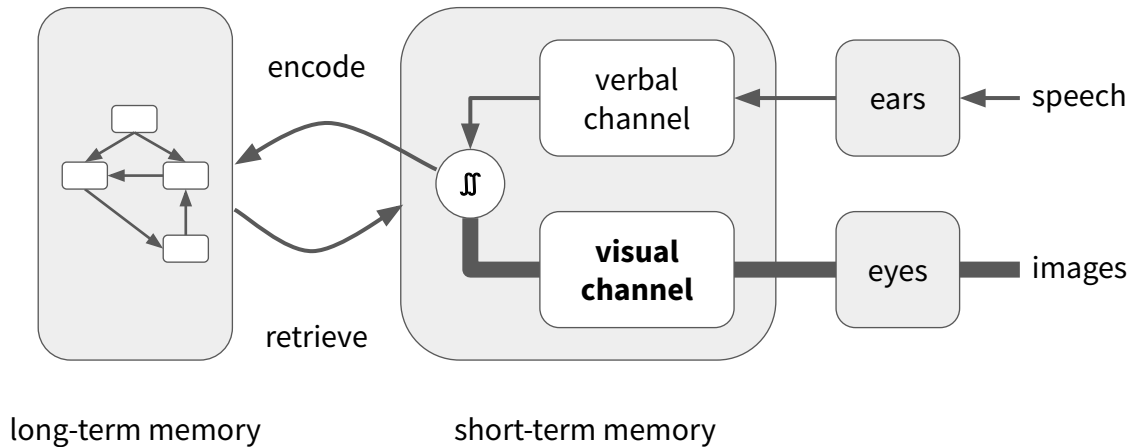
Integration



The last part of this picture is the work your brain does to *integrate* the information it's receiving through different channels. If you are hearing and seeing things at the same time, your brain tries to correlate them and store them together. Studies have shown that when images and words complement each other, the brain does a better job of remembering them both. The theory is that they are encoded together, so that later on, recall of one helps trigger recall of the other.

Description: the same diagram as before. The small boxes labeled "verbal channel" and "visual channel" are now connected to a small circle containing a double integral sign, and the arrow labeled "encode" that used to go from "short-term memory" to "long-term memory" now goes from the "integrate" circle to long-term memory. This is meant to show that verbal and visual information is integrated before being encoded and stored.

Unequal Weights



As we've drawn it so far, this diagram gives equal weight to the verbal and visual channels. In reality, most people get far more information, far more quickly, through their visual channel.

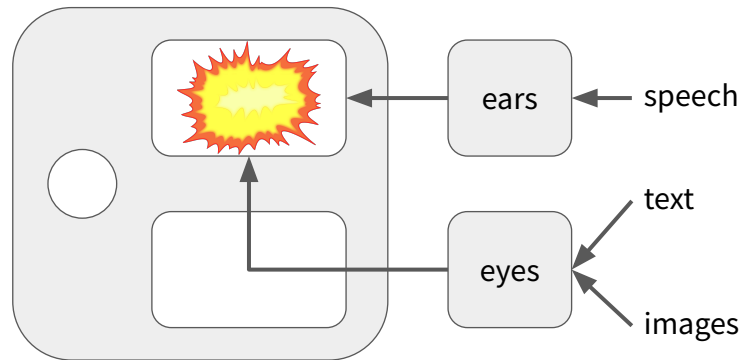
Description: the same diagram as before, but with the arrows from "images" to "eyes" to "visual channel" to "integrate" drawn much more heavily to emphasize them.

Unequal Weights

	Information	Neurons
Vision	83%	30%
Hearing	11%	2%
Smell	3.5%	8%
Touch	1.5%	
Taste	1%	

There's obviously a lot of variation between people, and our learner personas should remind us that not everyone can see and hear equally, but these figures give you an idea of how much information we get from different senses and how much of our brain is devoted to processing and storing it. What this tells us is that in most cases, our lessons should be as visual as possible.

Reading Is Weird



This model also explains why it's less effective to present the same information as text and speech, such as by reading text-heavy slides aloud or by having captions that "say" the same thing as the presenter. Text comes in through the visual channel, but is then shunted sideways to the verbal channel. Your brain can't help but try to reconcile the two streams of speech it's receiving (one through your eyes and one through your ears), and doing that takes mental effort away from encoding the information for storage.

Description: a subset of the memory and cognition diagram built up previously, showing short-term memory, the verbal and visual channels, and the eyes and ears. An arrow labeled "speech" goes into "ears", but there are now two arrows going into "eyes", one labeled "images" as before, and a new one labeled "text". There is also a new arrow going from the visual channel to the verbal channel, and an explosion icon in the box for the verbal channel.

Cognitive Load

The concept of “mental effort” turns out to be one of the most important in education. The technical term is *cognitive load*, and while the model continues to evolve, there are basically three kinds.

Description: this slide is blank other than its title.

Cognitive Load

How is her knee today?

To explain the differences, imagine that you're learning to speak Frisian, and that you've been asked to translate this sentence. You have to do two things: recall some words, and arrange them grammatically.

Description: this slide contains only the sentence "How is her knee today?"

Cognitive Load

How is her knee today?

hoe

har

is

HJOED

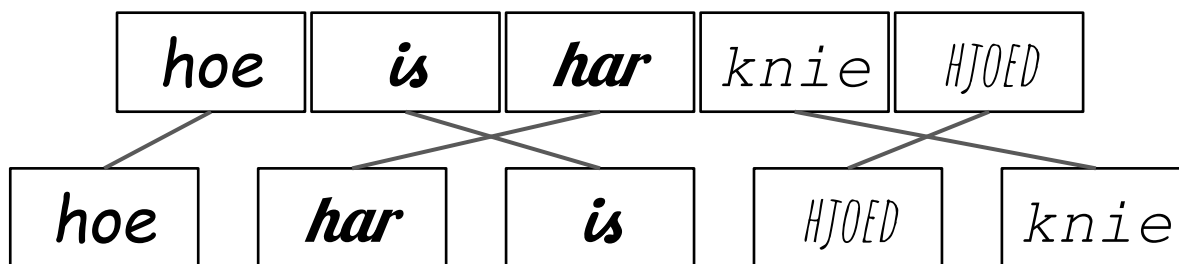
knie

But suppose I give you the words you need, so that all you have to do is arrange them in the right order.

Description: this slide contains the sentence and five boxes in a line labeled “hoe”, “har”, “is”, “hjoed”, and “knie”. Each label is in a different font.

Cognitive Load

How is her knee today?



I've made them problem much simpler by giving you the vocabulary and allowing you to focus your attention on the grammar.

Description: the five boxes have been rearranged to form the sentence "Hoe is har knie hjoed".

Cognitive Load

Intrinsic	What the learning task is supposed to be about	Grammar
Germane	Extra relevant work (such as retrieving supporting information)	Vocabulary
Extrinsic	Handling irrelevancies	Fonts

In general, intrinsic load is what the learning task is supposed to be about. Germane load is extra relevant work, such as remembering words or picking a proof strategy. Extrinsic load is irrelevancies. In this case, the *intrinsic* load is figuring out the order of the words. The *germane* load is recalling the words, which we can reduce by giving them to you. And the *extrinsic* load is the use of a different font for each word: it draws your attention, but it doesn't add anything to the lesson.

One Thing at a Time



Many kinds of learning use this technique. For example, children are learning to ride bicycles earlier than they were twenty years ago because of balance bikes (also called push bikes). These toys allow children to master balance and steering first, and then add pedalling.

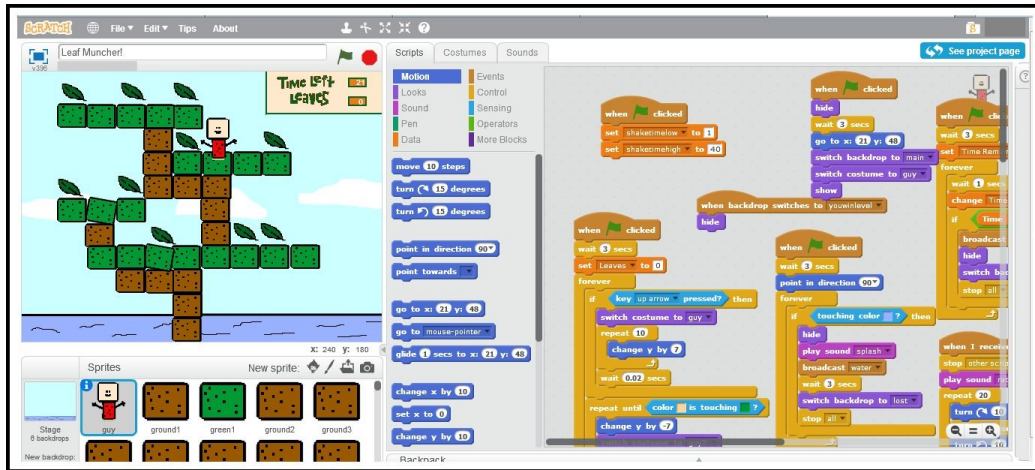
Image: child on balance bike

One Thing at a Time



Cognitive load theory explains why minimal guidance during instruction doesn't work—in other words, why throwing people into the deep end to solve “real” problems right from day one is less effective than explicitly teaching component skills one at a time. The reason is that combining two things is a separate skill in its own right. If you want to learn how to sing while playing the piano, you should learn how to play and learn how to sing and *then* combine the two. Once you've mastered the combined skill, you'll be able to do the two together from the start.

Scratch



Cognitive load theory also explains why blocks-based tools like Scratch are so effective. By taking away the possibility of syntax errors, they allow novices to focus on their programs' semantics.

Parsons Problems

Add all the positive numbers in a list

- 1) for each number in list
- 2) total := total + number
- 3) end for
- 4) if number > 0
- 5) end if
- 6) total := 0

Problems where learners are given the pieces to rearrange are called Parsons Problems, and have been shown to be more effective in early stages of learning programming than writing code from scratch.

Description: The instruction to add all the positive numbers in a list, and the six lines of code, are on the left.

Parsons Problems

Add all the positive numbers in a list

- | | |
|----------------------------|----------------------------|
| 1) for each number in list | 6) total := 0 |
| 2) total := total + number | 1) for each number in list |
| 3) end for | 4) if number > 0 |
| 4) if number > 0 | 2) total := total + number |
| 5) end if | 5) end if |
| 6) total := 0 | 3) end for |

Problems where learners are given the pieces to rearrange are called Parsons Problems, and have been shown to be more effective in early stages of learning programming than writing code from scratch.

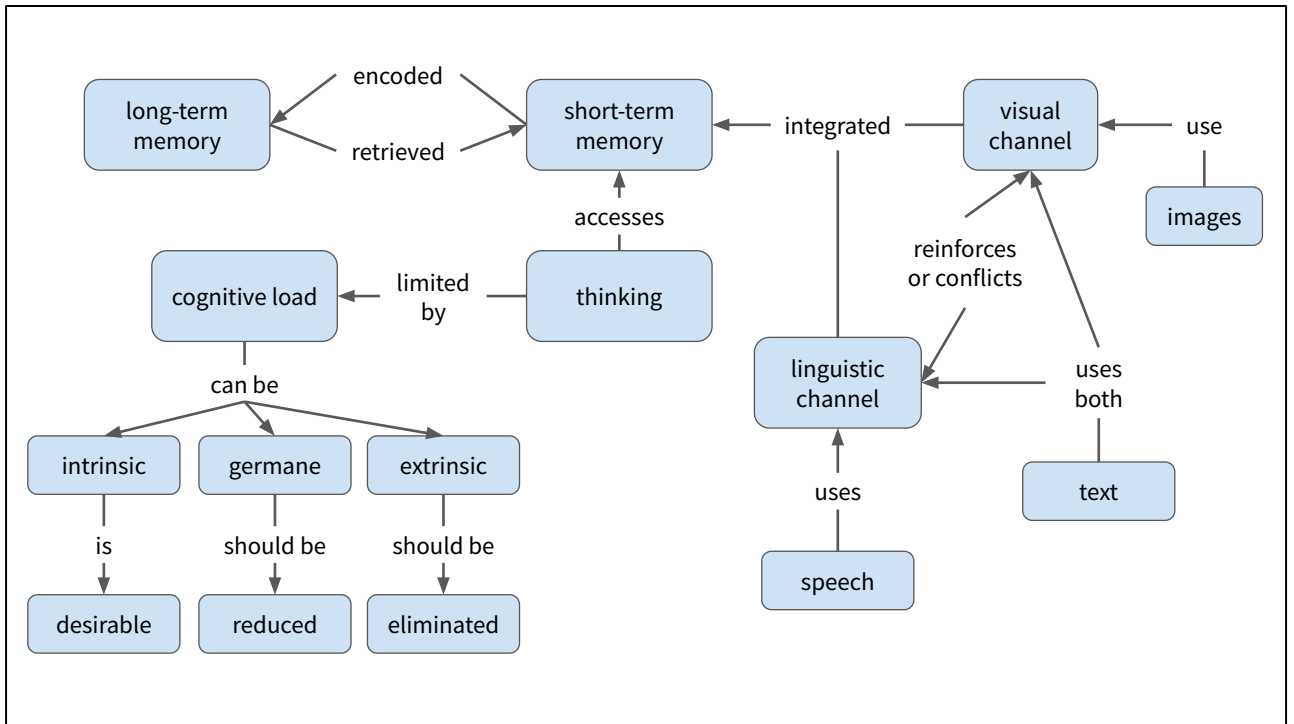
Description: this slide has the same description and lines of code on the left. On the right, the six lines of code have been rearranged into the correct order.

Exercise

Create a Parsons Problem for a simple task with 4-6 steps. Do not show the correct order: we will solve some of your problems in pairs.



Create a Parsons Problem for a simple programming task and try it out with your neighbor.



Concept map for cognitive load.

Descriptive text: a concept map summarizing this lesson.